Md Kamruzzaman

Backend & MLOps Engineer | Python | Django | FastAPI
Dhaka, Bangladesh (UTC+6) | +880 1710 274982 | kamruzzamanmoon13@gmail.com
linkedin.com/in/kamruzzaman-moon-5a83281b7 | github.com/moon121314 | moon121314.github.io

SUMMARY

B.Tech Computer Science graduate (Machine Learning, 2025) and Software Engineering Intern at Felna Tech. Builds secure Python backend APIs and ML serving pipelines using Django, FastAPI, PostgreSQL, Docker, and MLflow. Seeking a Backend or MLOps engineering role.

EXPERIENCE

Software Engineering Intern | *Felna Tech* Oct 2026 – Present

- Build and maintain REST APIs and backend services in Python with Django and FastAPI.
- Integrate PostgreSQL, optimize complex queries, and enforce data validation.
- Participate in agile meetings and code reviews; troubleshoot production issues.

PROJECTS

Factory Traffic Management System | *FastAPI, PostgreSQL, SQLAlchemy, Pytest, Docker* | Live API · GitHub Oct 2026

- Built a safety-critical, event-driven traffic backend with state machines and emergency preemption logic.
- Prevented race conditions with PostgreSQL row-level locking (`SELECT ... FOR UPDATE`) on concurrent sensor events.
- Added two-pass crash recovery and 50 automated tests, including a concurrency stress test.

ML Expense Platform | *FastAPI, scikit-learn, MLflow, Docker, Kubernetes, Prometheus, Grafana* | Live API · GitHub Sep – Oct 2026

- Built an end-to-end ML serving pipeline that classifies free-text expenses (TF-IDF + Logistic Regression), with experiments tracked in MLflow.
- Reached **~5 ms inference latency** by loading the model at startup; load-tested at **500+ requests/second**.
- Deployed on Kubernetes with Prometheus metrics, Grafana dashboards, and Evidently data-drift detection.

Docu-RAG API | *FastAPI, LangChain, ChromaDB, Google Gemini API, Docker* | GitHub Oct 2026 – Present

- Built a RAG API for Q&A over PDFs with **2–4 second response latency**.
- Created a LangChain ingestion pipeline storing embeddings in ChromaDB; grounded Gemini answers in retrieved context to reduce hallucinations.

Expense Tracker API | *FastAPI, PostgreSQL, SQLAlchemy, Docker, GitHub Actions* | Live API · GitHub Mar – Oct 2026

- Built a REST API with 8 endpoints, JWT authentication, user-scoped data isolation, filtering, pagination, and strict input validation.
- Reached **91% test coverage** with pytest; automated CI with GitHub Actions.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, Bash
- **Backend:** Django, FastAPI, REST APIs, SQLAlchemy, Pydantic, Express.js, JWT, RBAC
- **Databases:** PostgreSQL, MongoDB, ChromaDB
- **AI & MLOps:** MLflow, scikit-learn, LangChain, Google Gemini API, RAG, Kubernetes
- **DevOps:** Docker, GitHub Actions (CI/CD), Prometheus, Grafana, Render, Vercel
- **Core:** System Design, Data Structures, Algorithms, OOP

EDUCATION

B.Tech, Computer Science & Engineering (Machine Learning), Lovely Professional University 2025

CERTIFICATIONS

Meta Back-End Developer (Coursera) | Natural Language Processing Specialization (DeepLearning.AI)

GITHUB ACTIVITY

143 contributions in the past year (about 95% commits) across 14 repositories; pull-request workflow with CI. github.com/moon121314